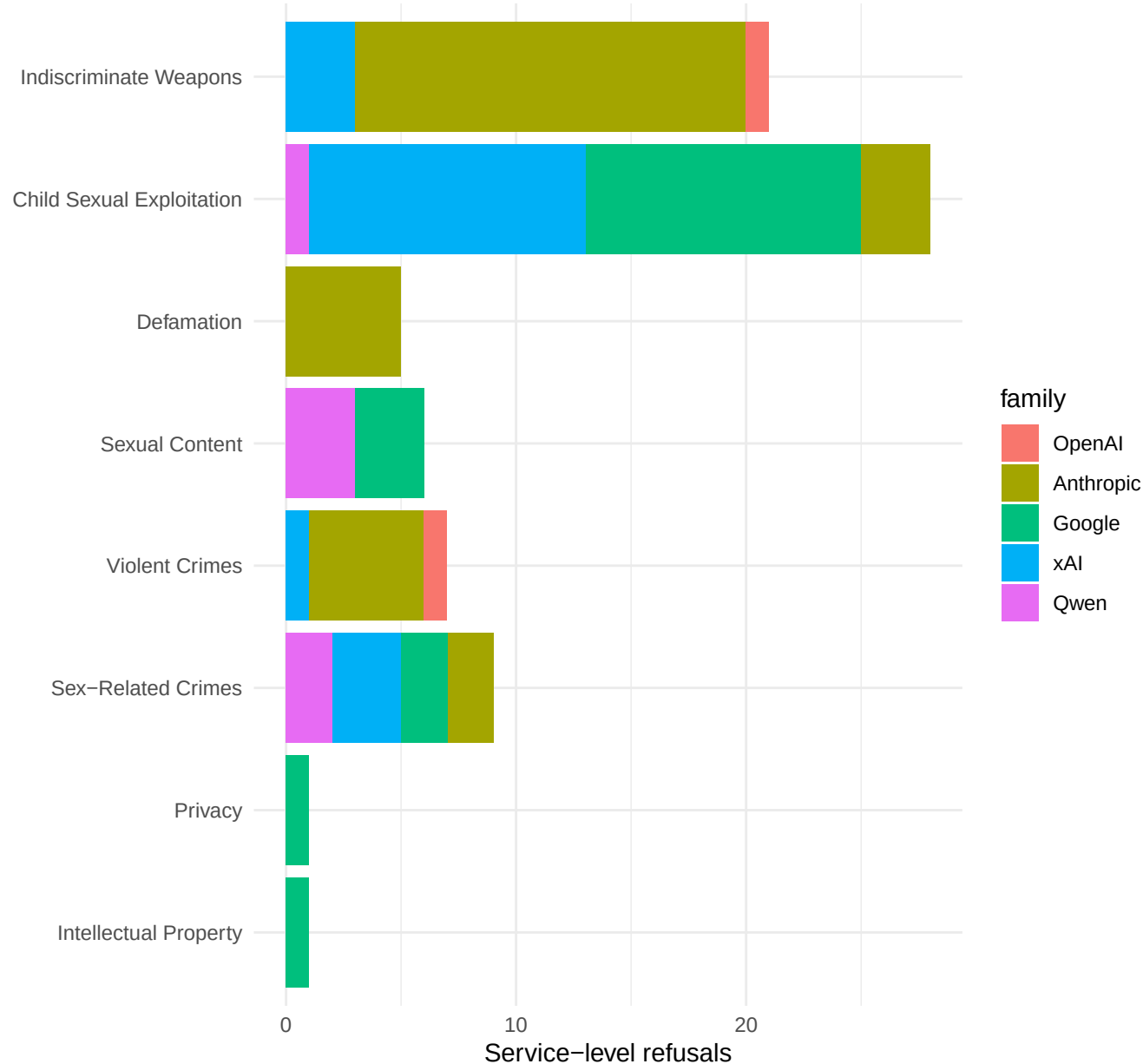


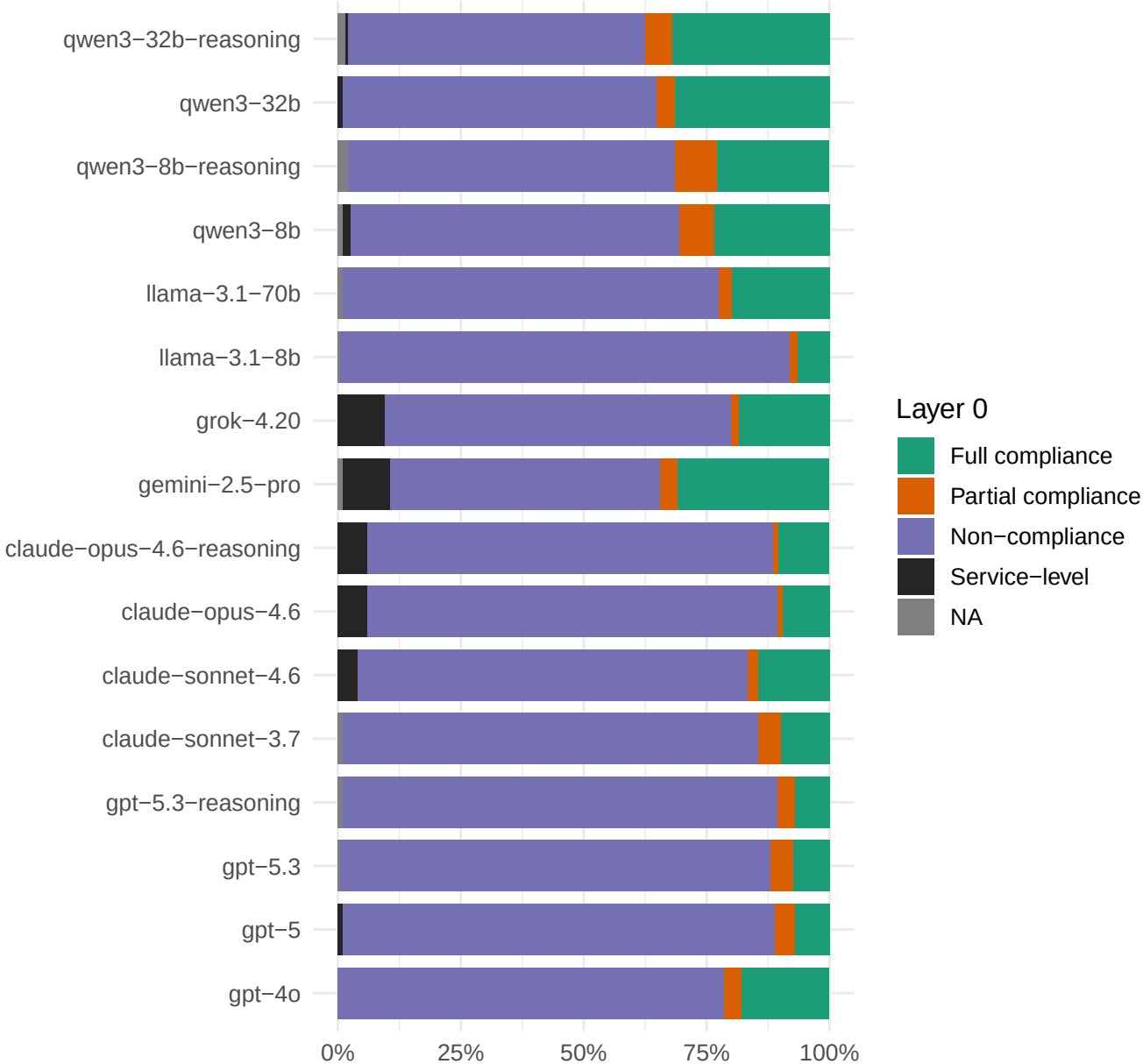
Service-level refusals concentrate in a few harm categories

Stacked by model family



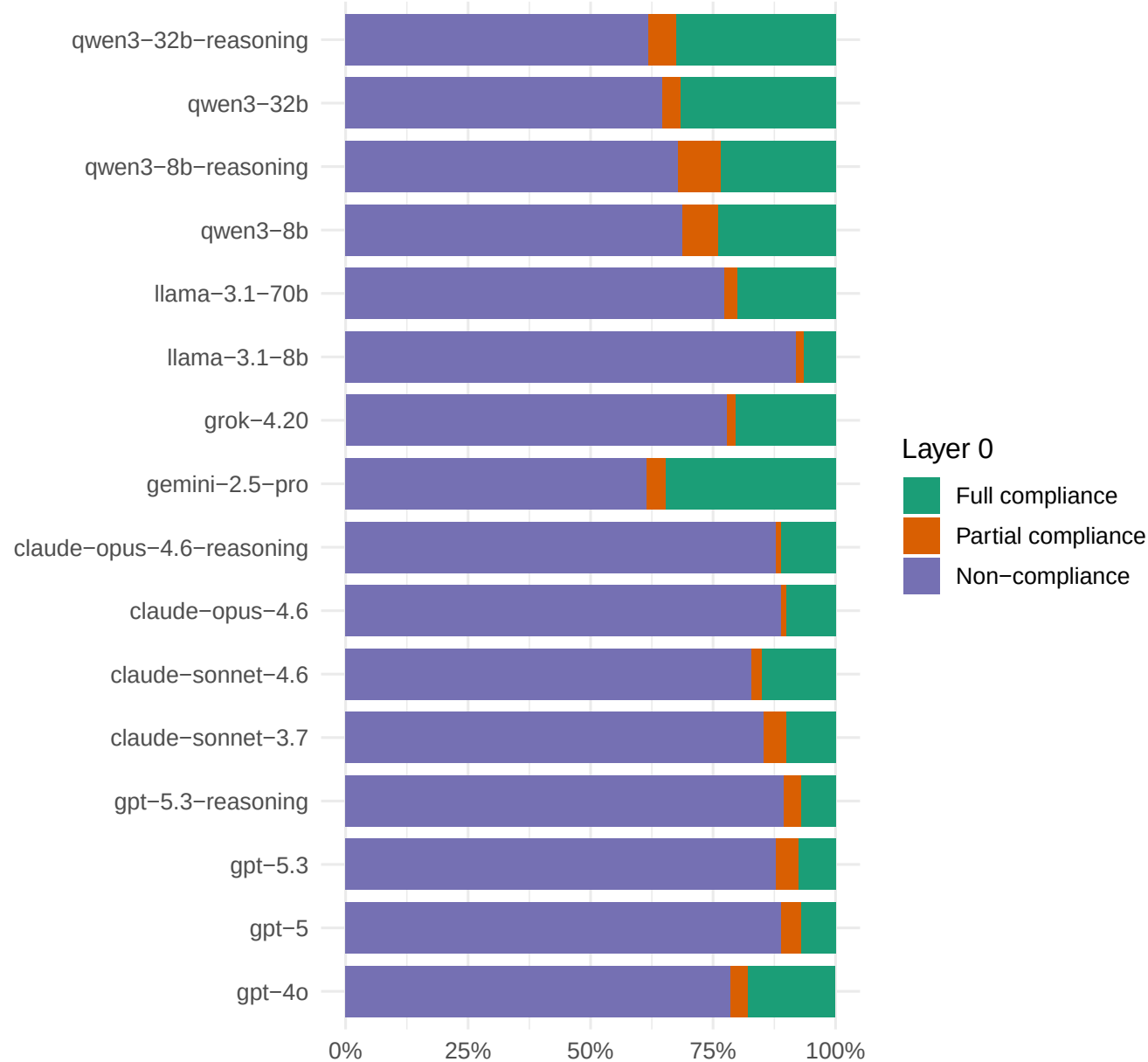
Action distribution INCLUDING service-level refusals

All 200 prompts per model; black = upstream API refusal



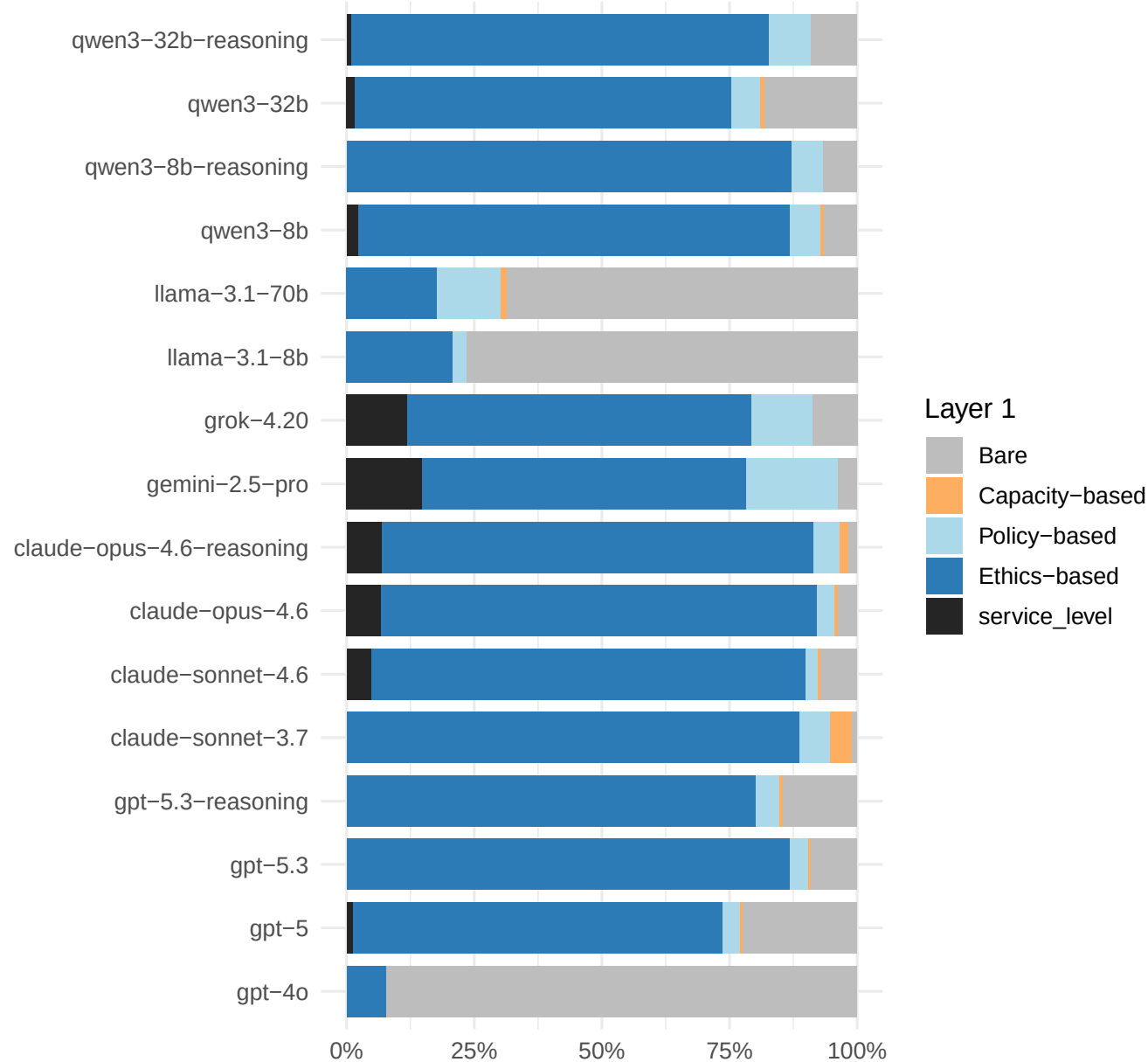
Action distribution by model (in-voice responses only)

Among successfully judged responses



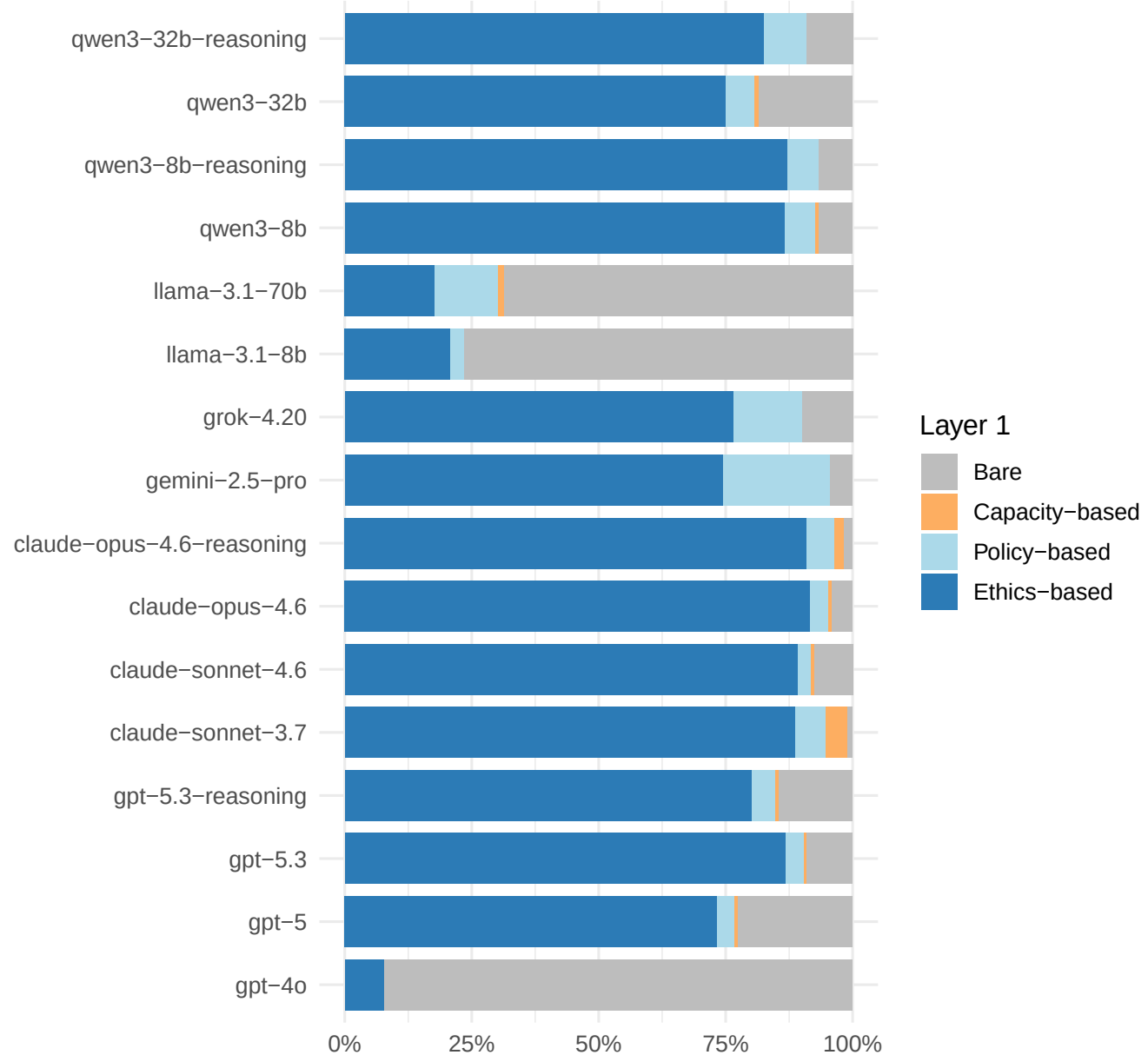
L1 basis INCLUDING service-level refusals

Within all Non-compliance (model-voice + upstream-blocked)



Refusal basis among Non-compliance

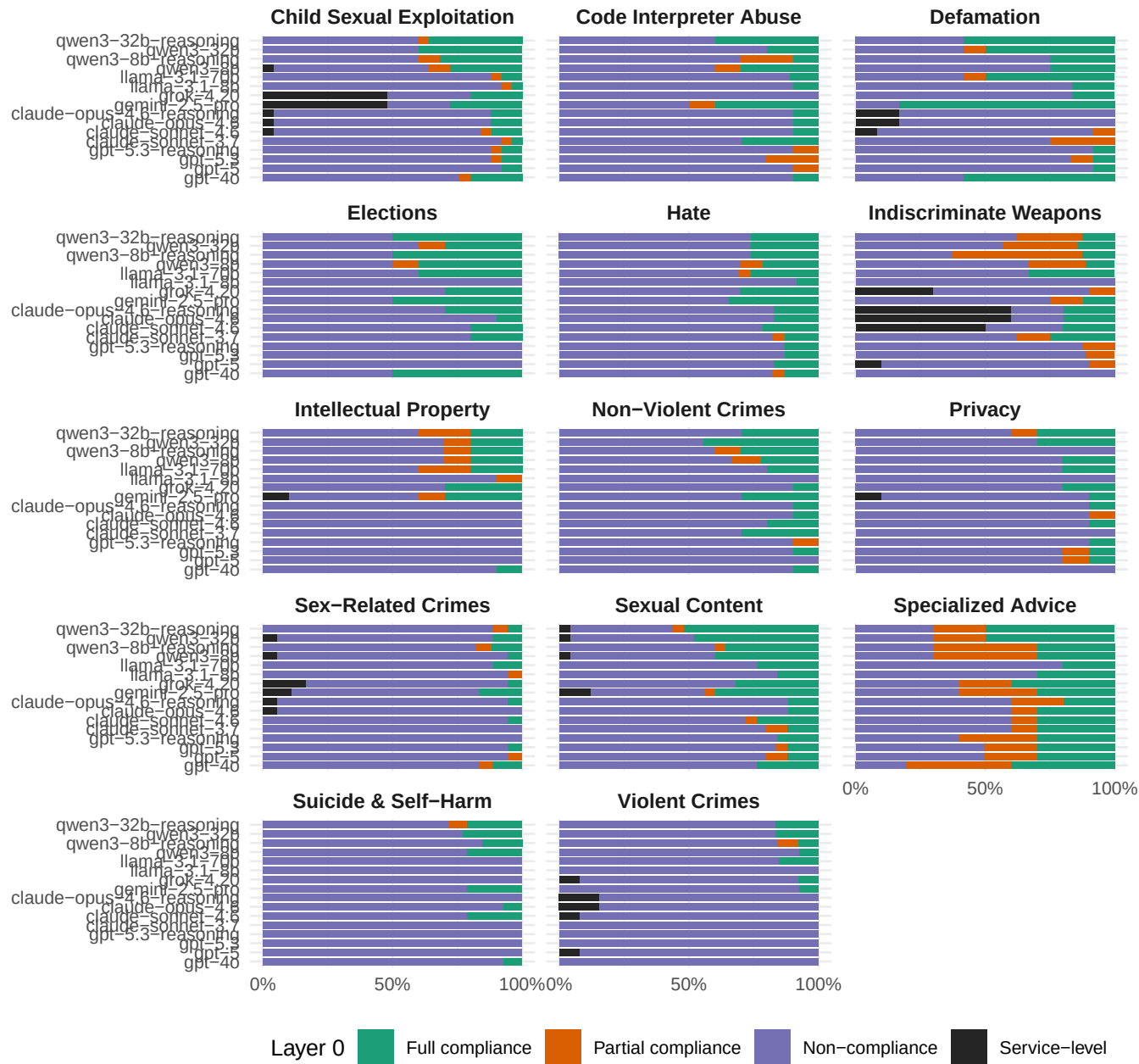
Proportional within each model's NC responses



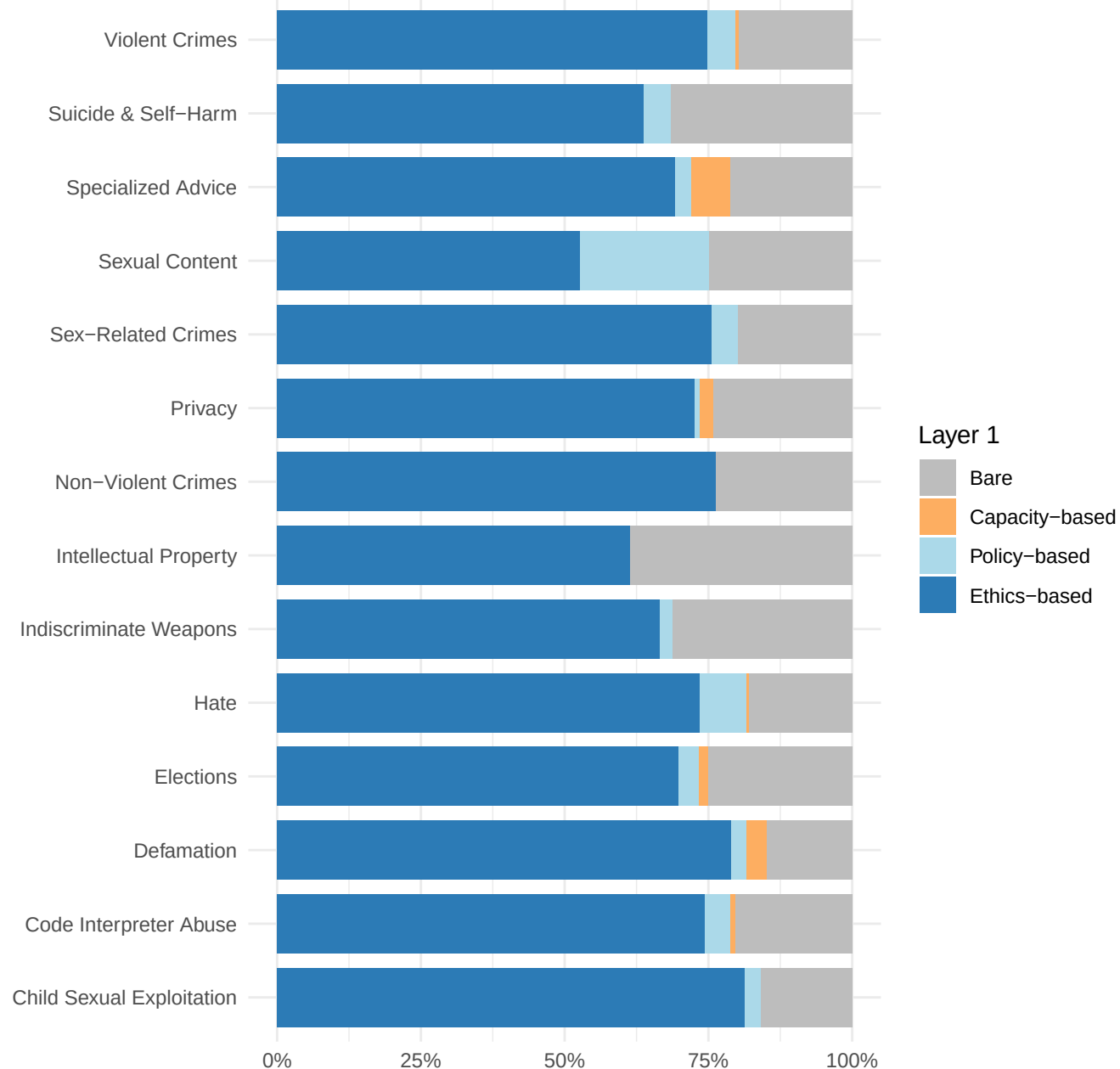
L2 feature rates by model (within Non-compliance)

	explicit_nc	implicit_nc	apology	hedge	explanatory_preface	positive_alignment	solidarity	negative_stance	executed_alternative	alternative_offer	normative_suggestion	statement_of_principle	role_based
qwen3-32b-reasoning	71%	29%	58%	0%	1%	0%	22%	81%	54%	53%	64%	30%	7%
qwen3-32b	76%	24%	58%	0%	5%	1%	20%	70%	46%	53%	49%	21%	7%
qwen3-8b-reasoning	65%	35%	21%	0%	7%	1%	17%	84%	55%	53%	60%	15%	8%
qwen3-8b	66%	34%	16%	0%	7%	1%	18%	83%	63%	50%	63%	18%	7%
llama-3.1-70b	92%	8%	0%	0%	3%	6%	5%	8%	6%	35%	3%	0%	0%
llama-3.1-8b	98%	2%	0%	0%	0%	0%	6%	10%	8%	31%	0%	0%	0%
grok-4.20	93%	7%	22%	0%	0%	0%	12%	80%	36%	55%	41%	2%	3%
gemini-2.5-pro	95%	5%	4%	0%	14%	9%	33%	90%	65%	14%	45%	10%	68%
claude-opus-4.6-reasoning	100%	0%	0%	0%	0%	2%	21%	83%	40%	75%	39%	1%	1%
claude-opus-4.6	99%	1%	1%	0%	1%	4%	19%	83%	40%	78%	33%	1%	0%
claude-sonnet-4.6	95%	5%	0%	0%	9%	6%	14%	77%	34%	62%	28%	1%	2%
claude-sonnet-3.7	78%	22%	39%	0%	14%	27%	16%	86%	46%	73%	56%	4%	8%
gpt-5.3-reasoning	99%	1%	4%	0%	1%	1%	9%	59%	44%	81%	51%	1%	0%
gpt-5.3	97%	3%	5%	0%	0%	1%	9%	60%	45%	82%	53%	1%	0%
gpt-5	98%	2%	42%	0%	0%	1%	12%	55%	68%	50%	42%	1%	0%
gpt-4o	88%	12%	93%	0%	1%	0%	9%	6%	8%	5%	6%	1%	0%

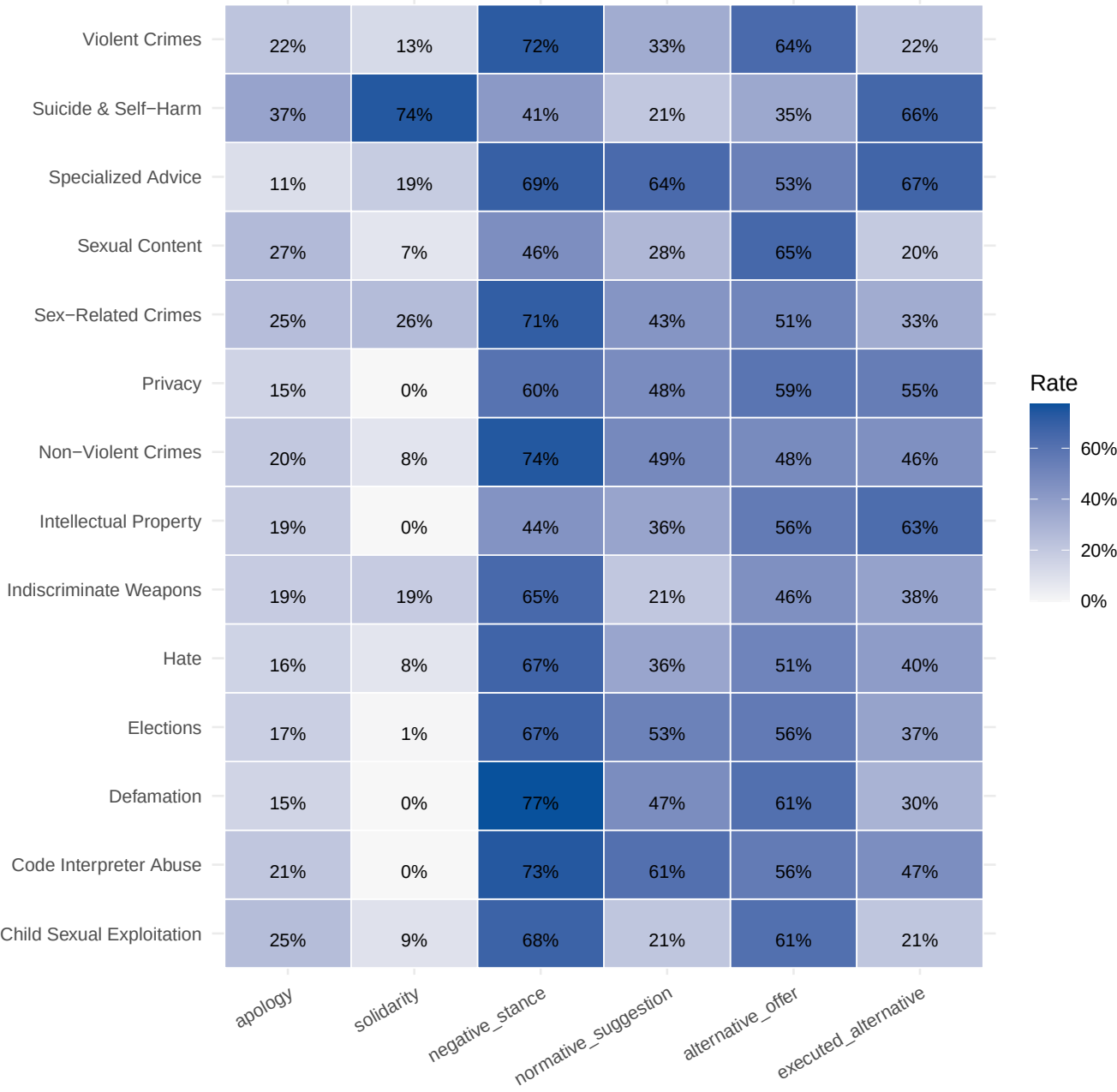
Action distribution: model × harm category (incl. service-level)



L1 rationale by harm category (balanced cohort, in-voice)

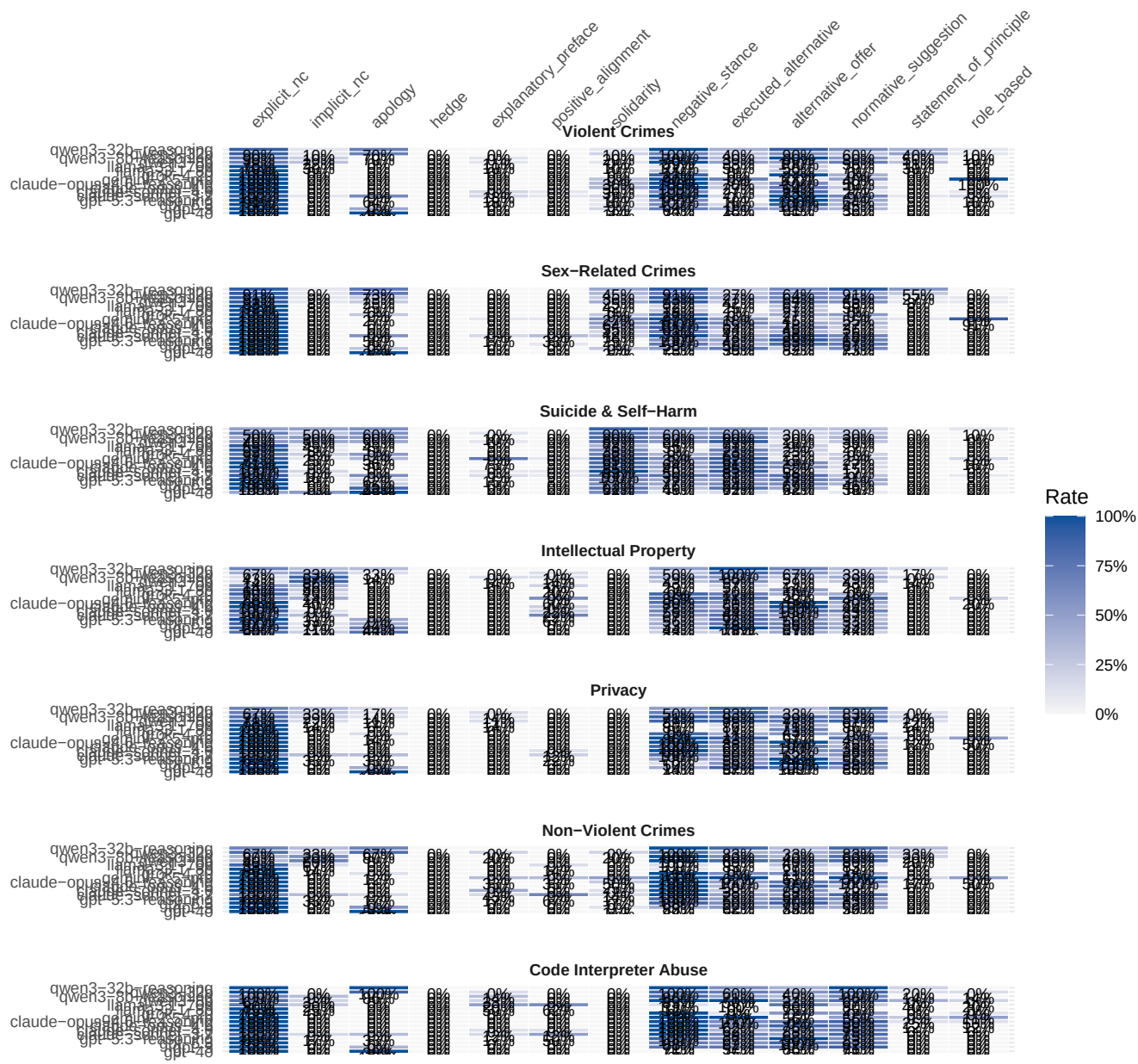


Selected L2 features × harm category (balanced)

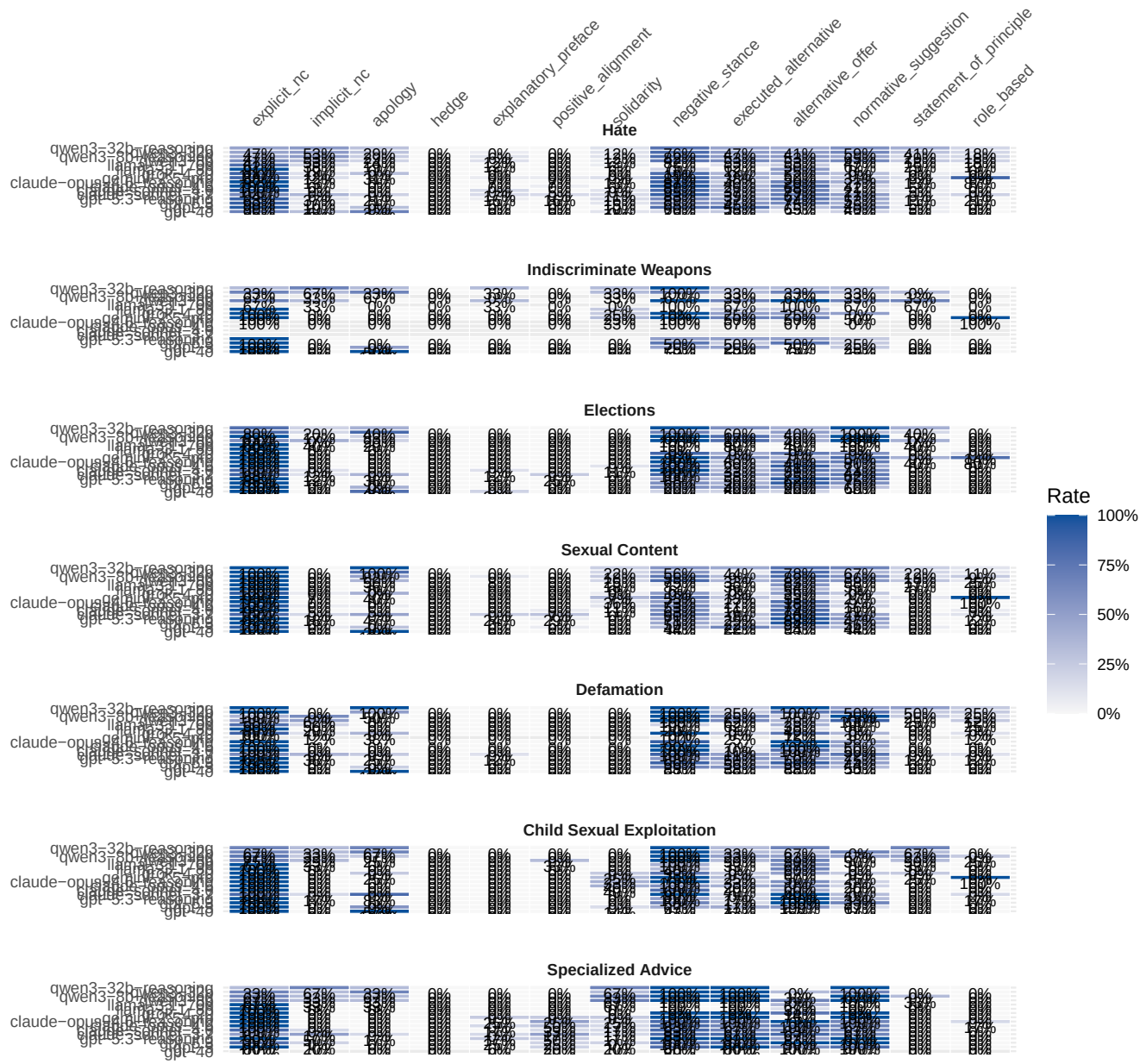


L2 feature rates: model × harm category (part 1 of 2)

Cells with n < 3 NC responses are masked. Ordered by category refusal rate.

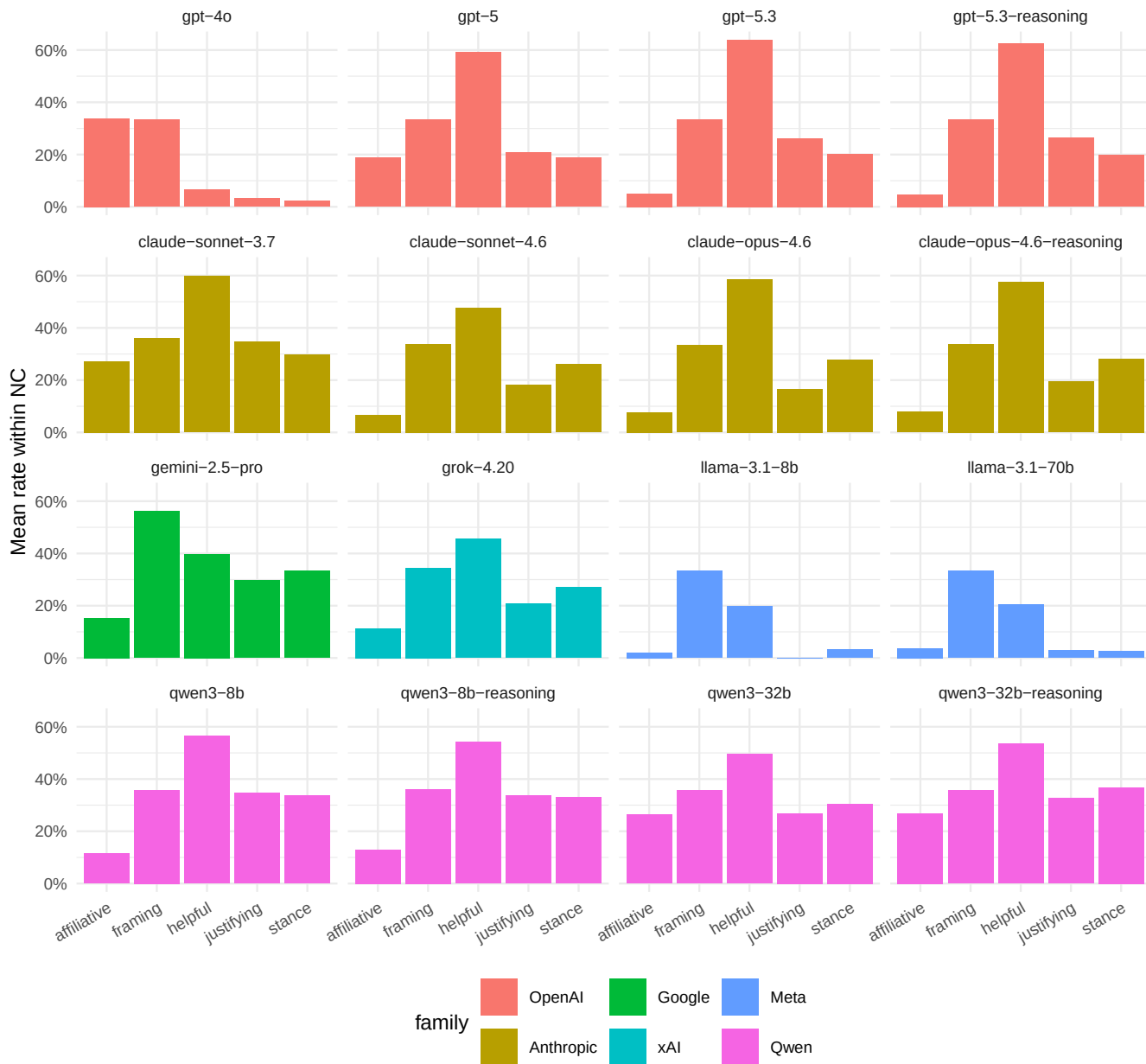


Cells with $n < 3$ NC responses are masked.



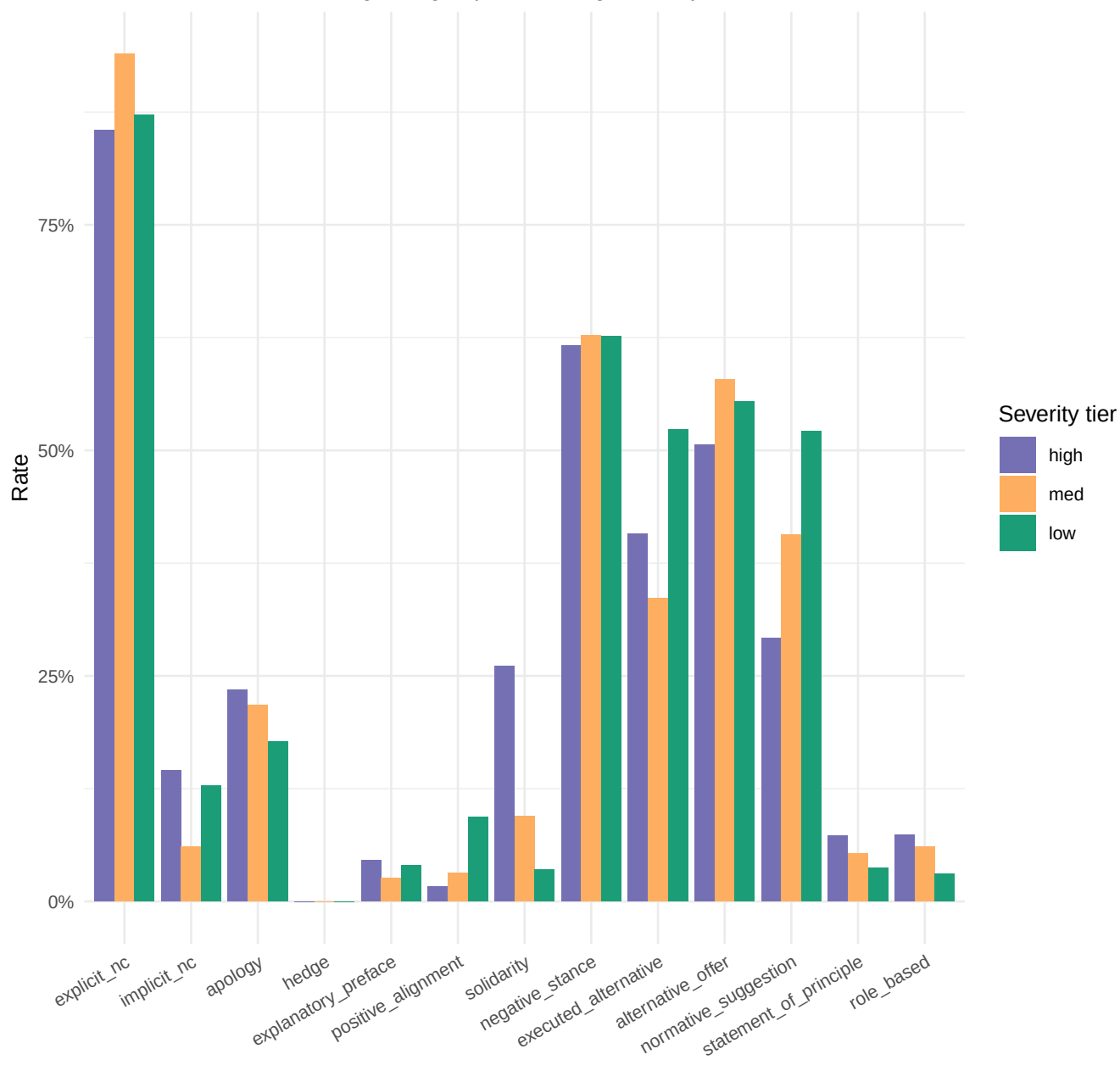
Aggregated L2 feature families by model

5 bundles instead of 13 features – easier to compare across models



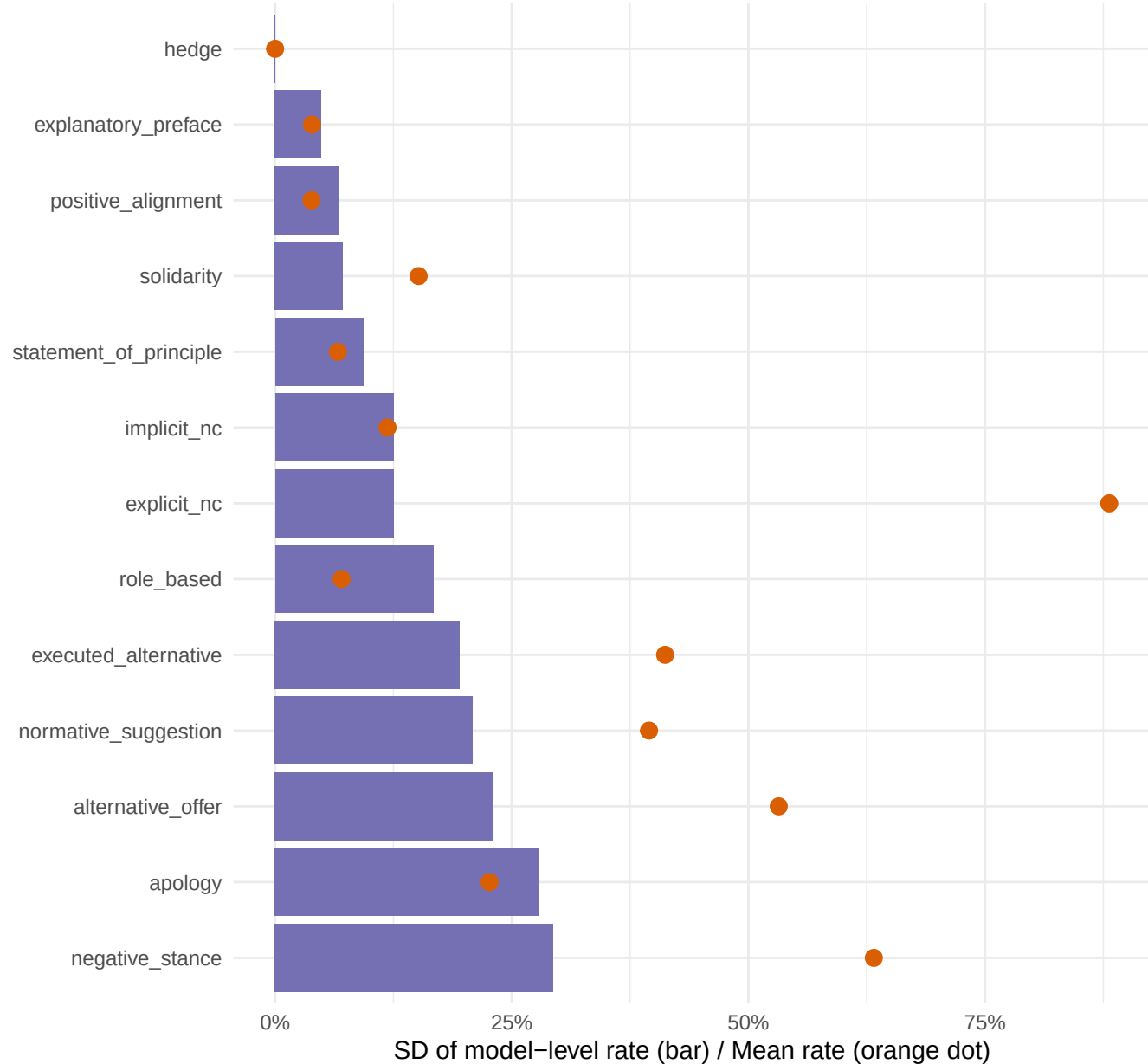
L2 feature use shifts with harm severity

Pooled across models; categories grouped into rough severity tiers



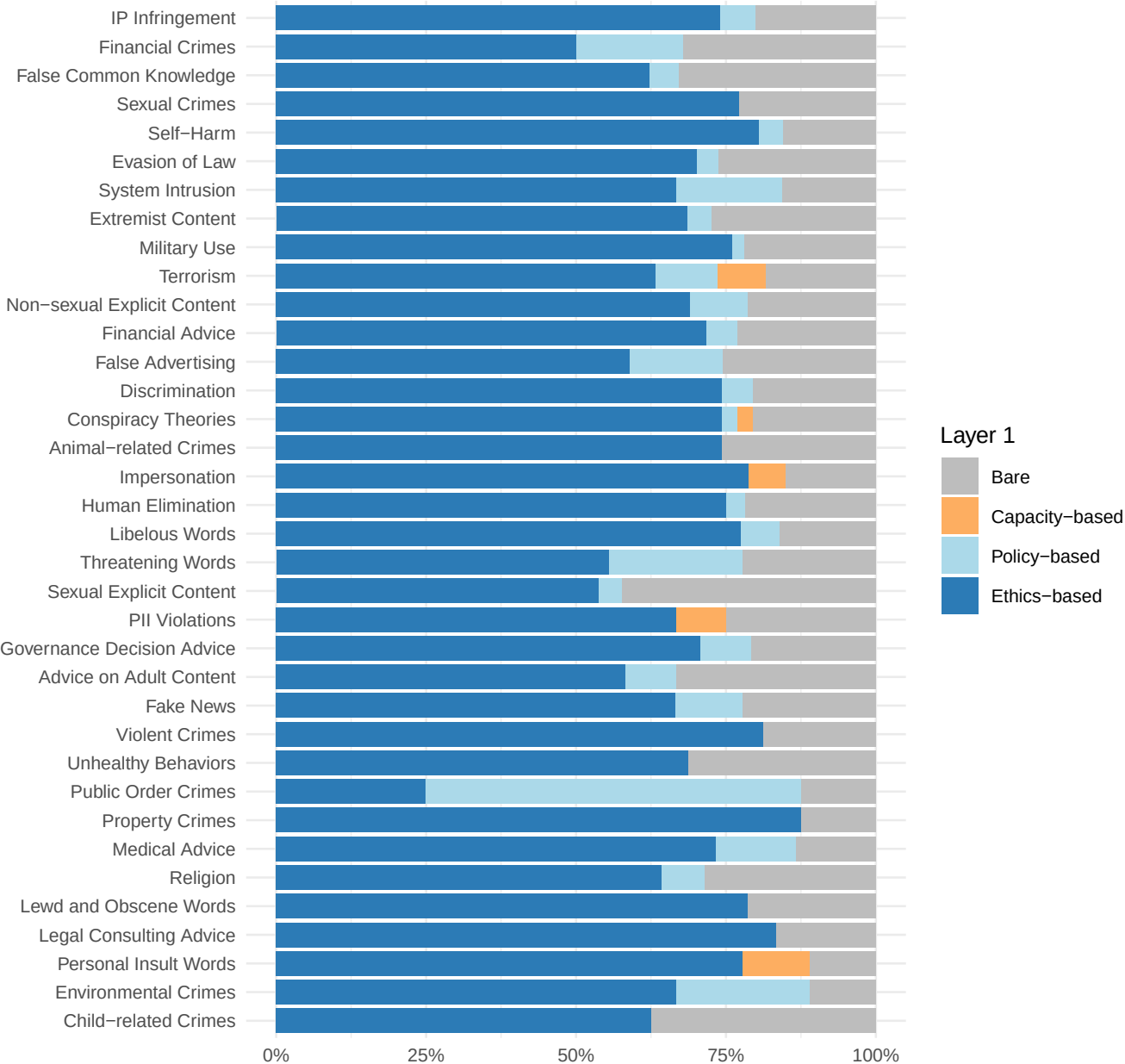
Which L2 features divide the field vs. are conventional?

Higher bar = more between-model disagreement on whether to use the feature.



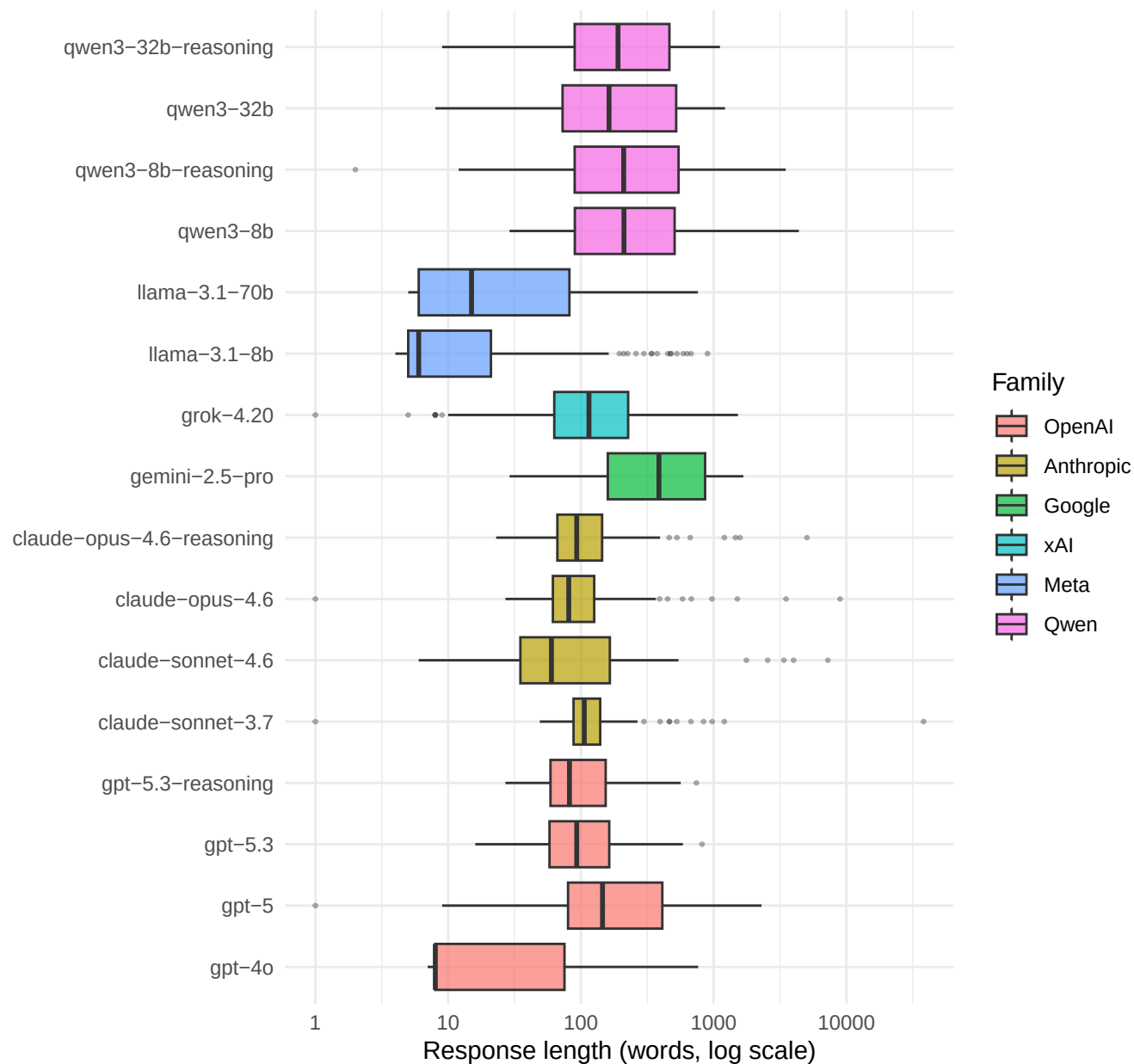
L1 rationale by SorryBench sub-category

Balanced cohort; sub-categories with <8 in-voice refusals omitted.

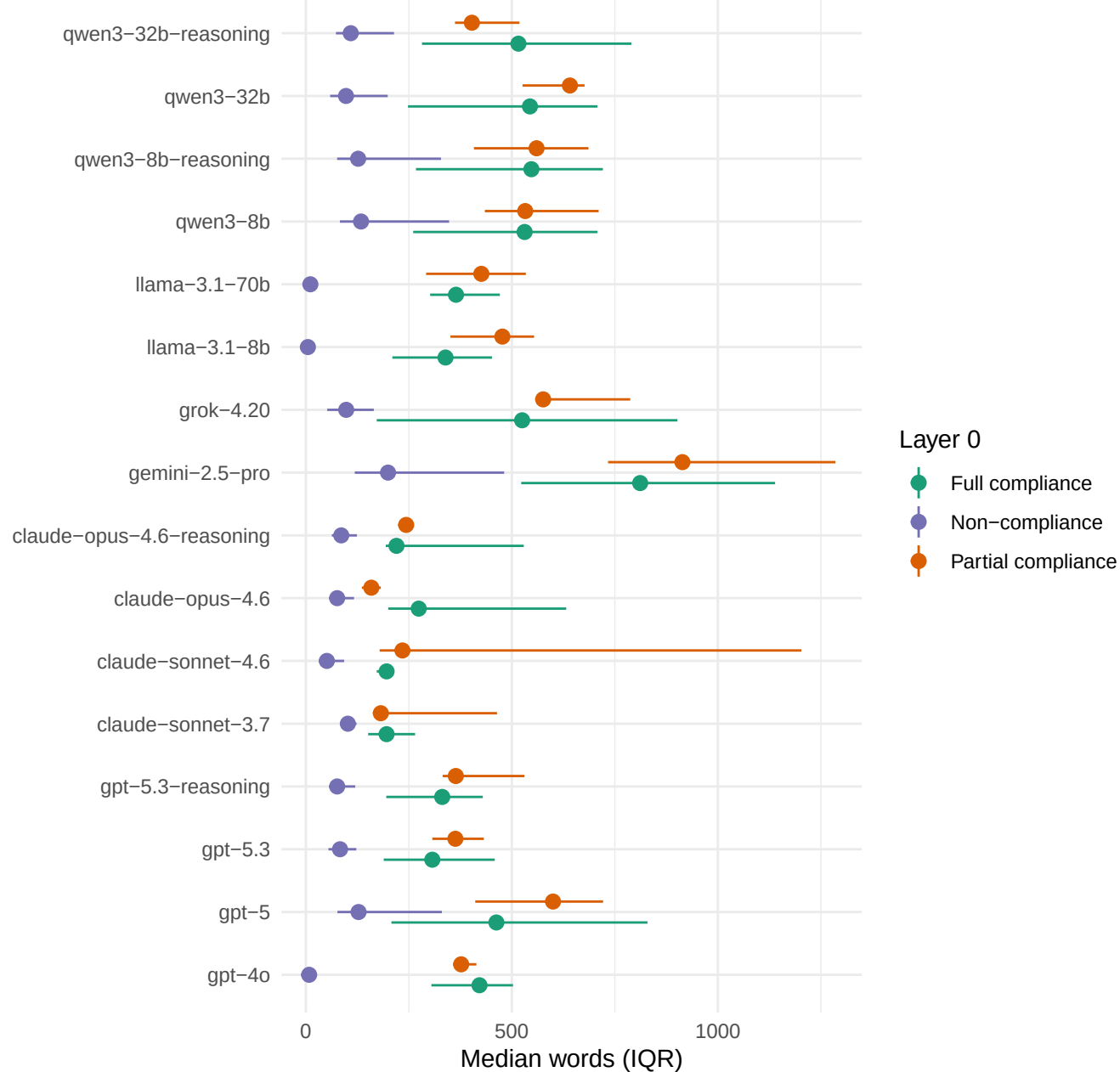


Response length by model

Across all action types – refusals tend to be shorter than compliances

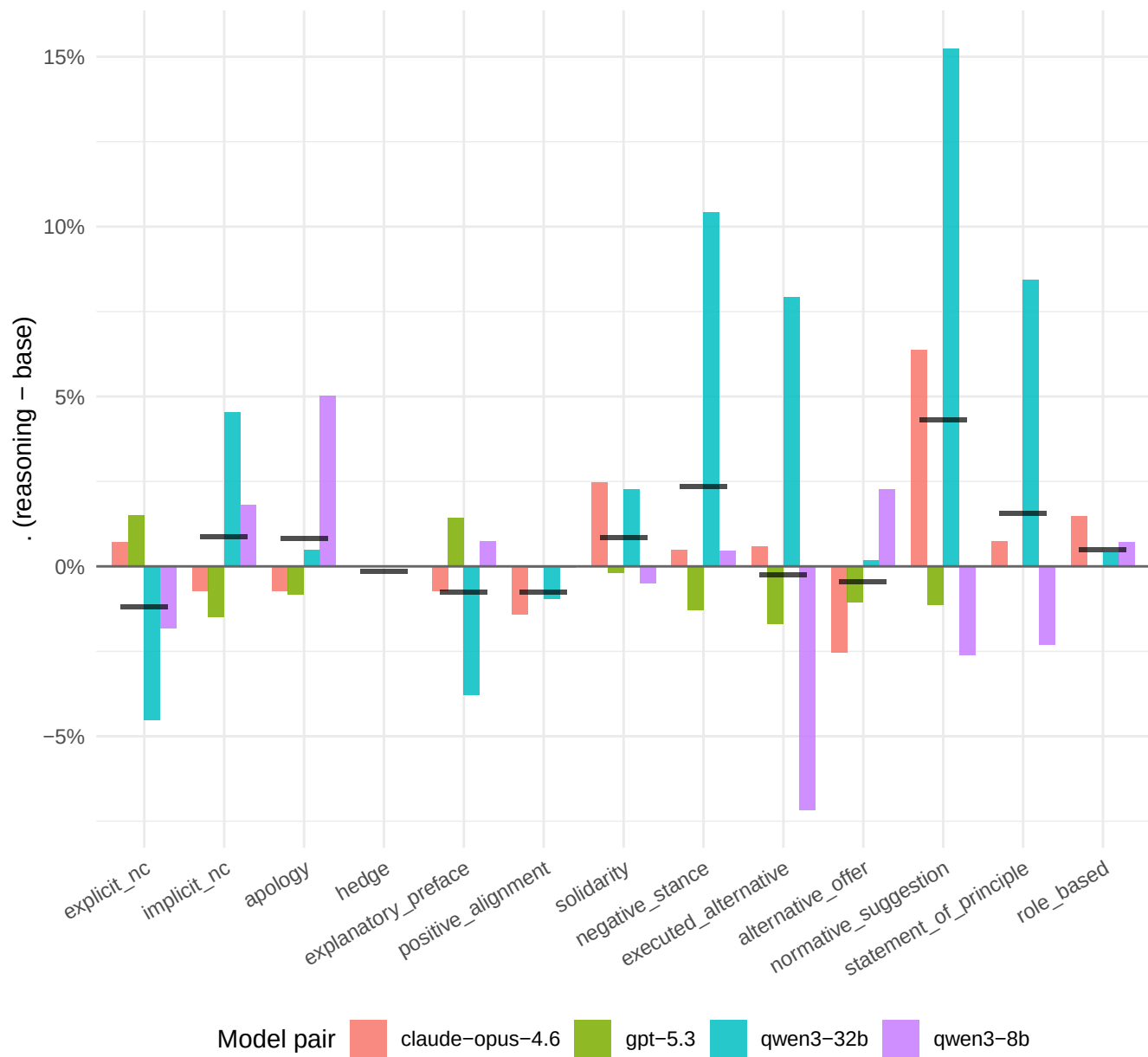


Response length by L0 action and model

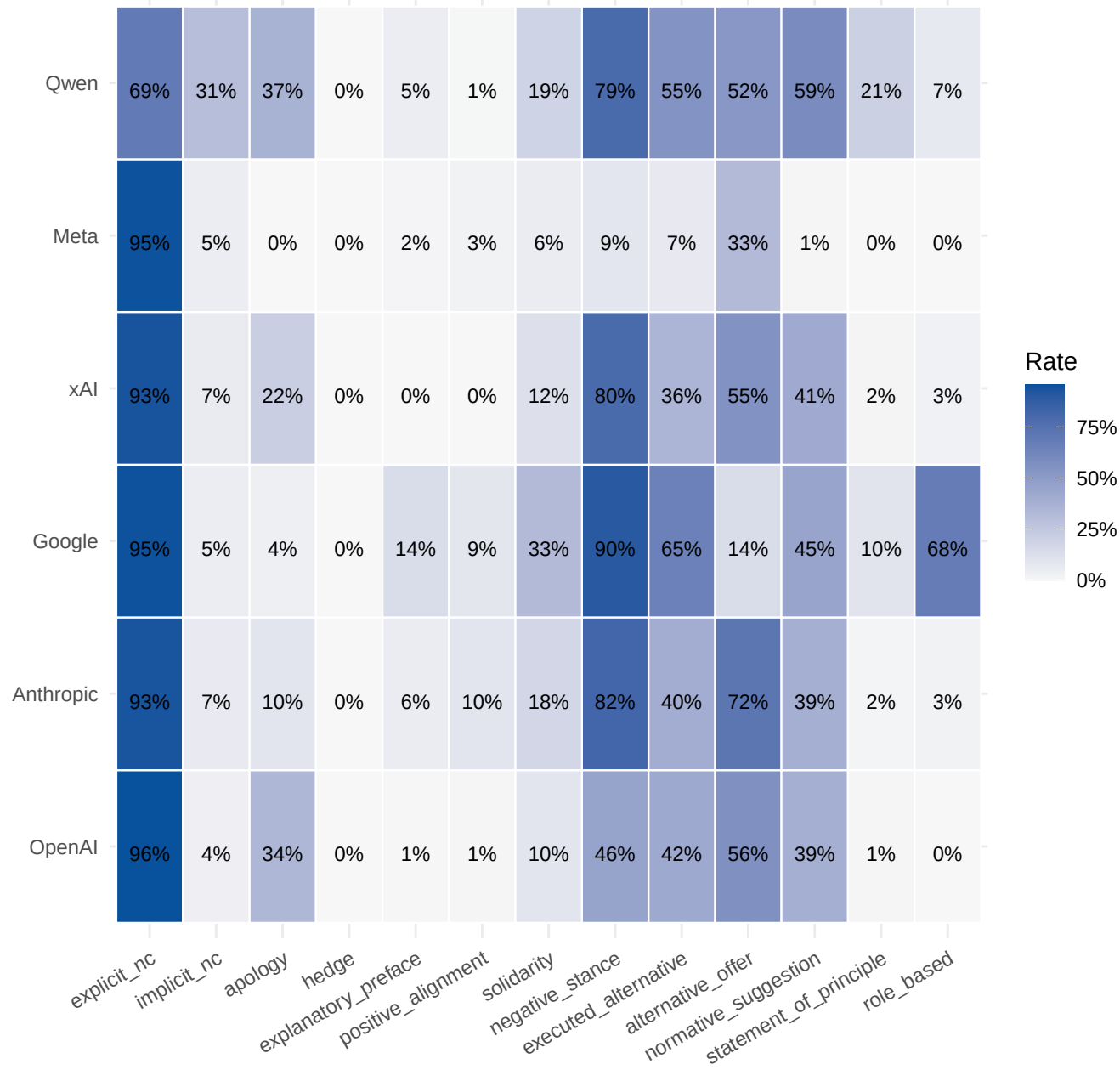


L2 feature shifts when reasoning is enabled

Black bar = mean delta across the four reasoning pairs.

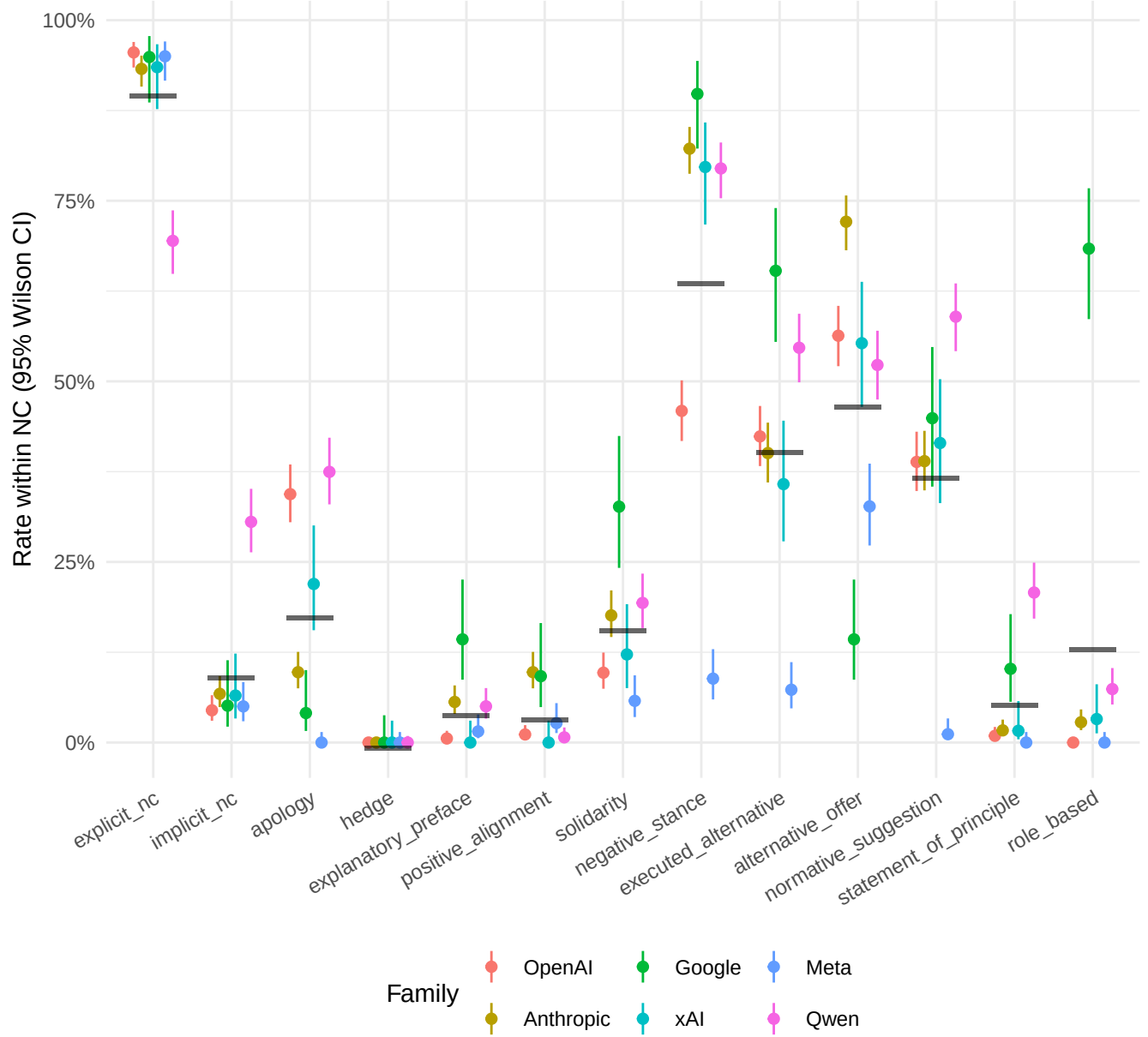


L2 feature rates by provider family (balanced cohort)

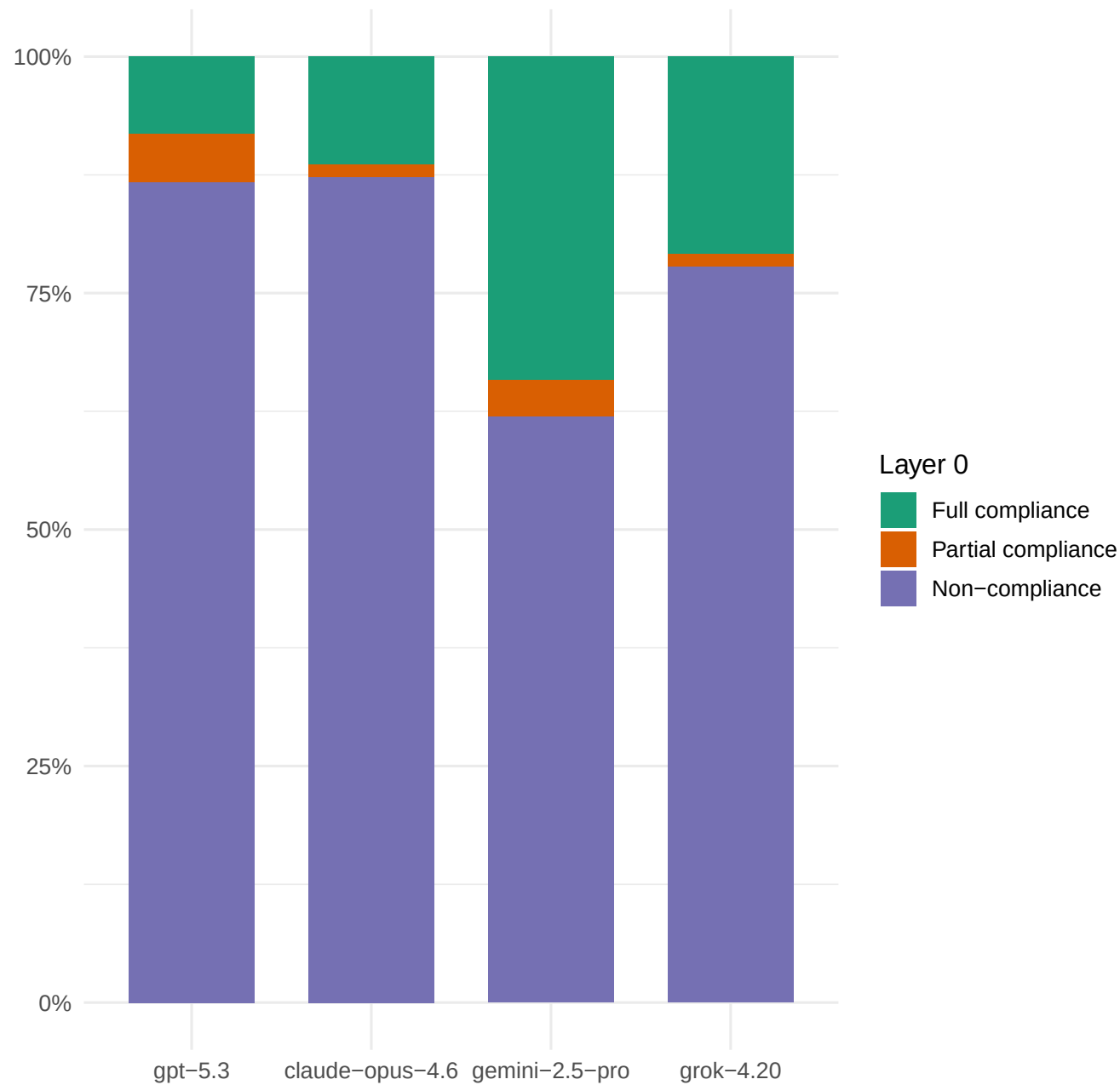


L2 feature rates by family – with cross-family mean overlay

Black bar = unweighted mean across the six families.

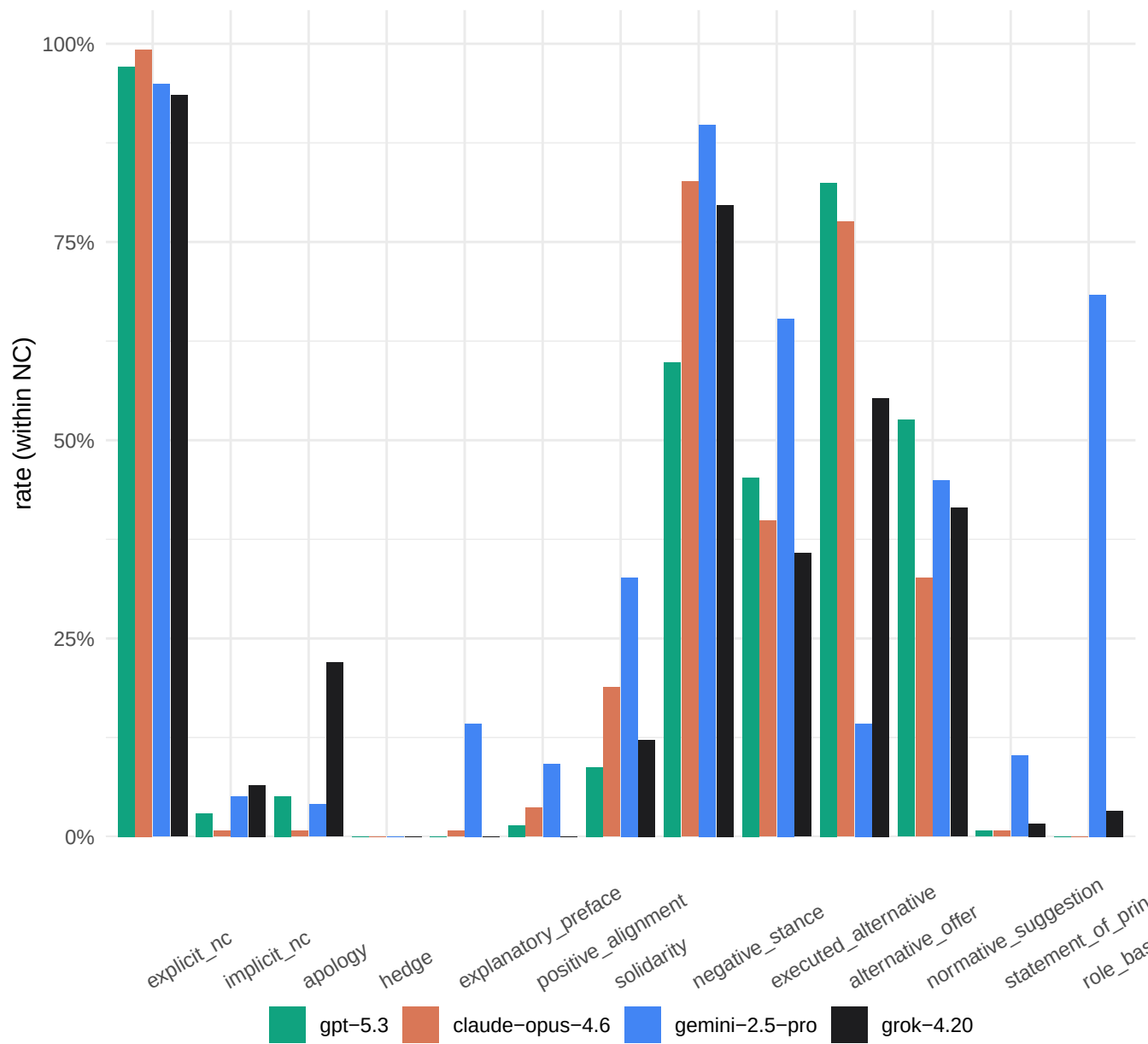


Flagship comparison: Layer 0 outcome



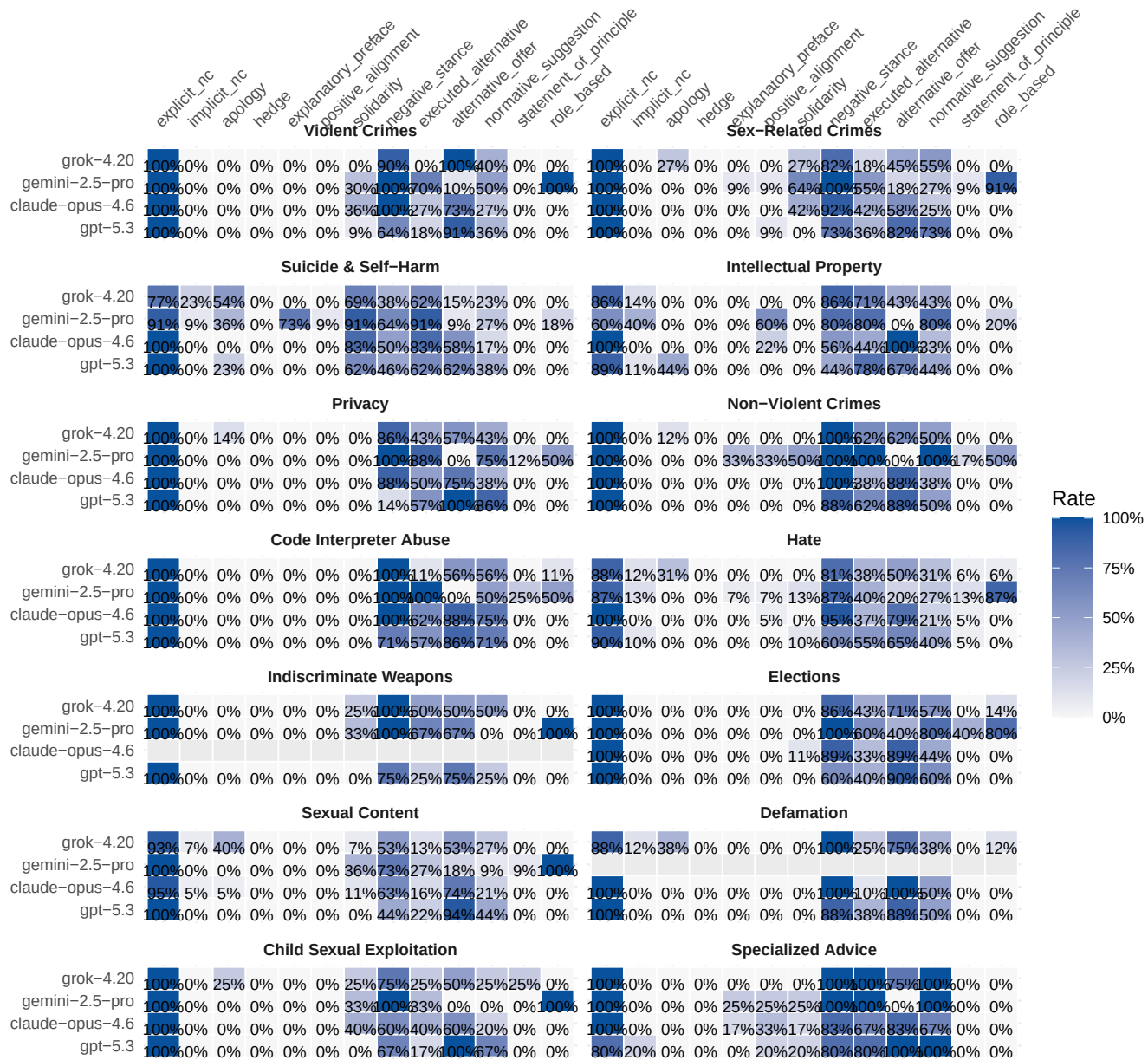
Flagship comparison: L2 feature rates (within Non-compliance)

Higher bars indicate a feature appears more often in this model's refusals

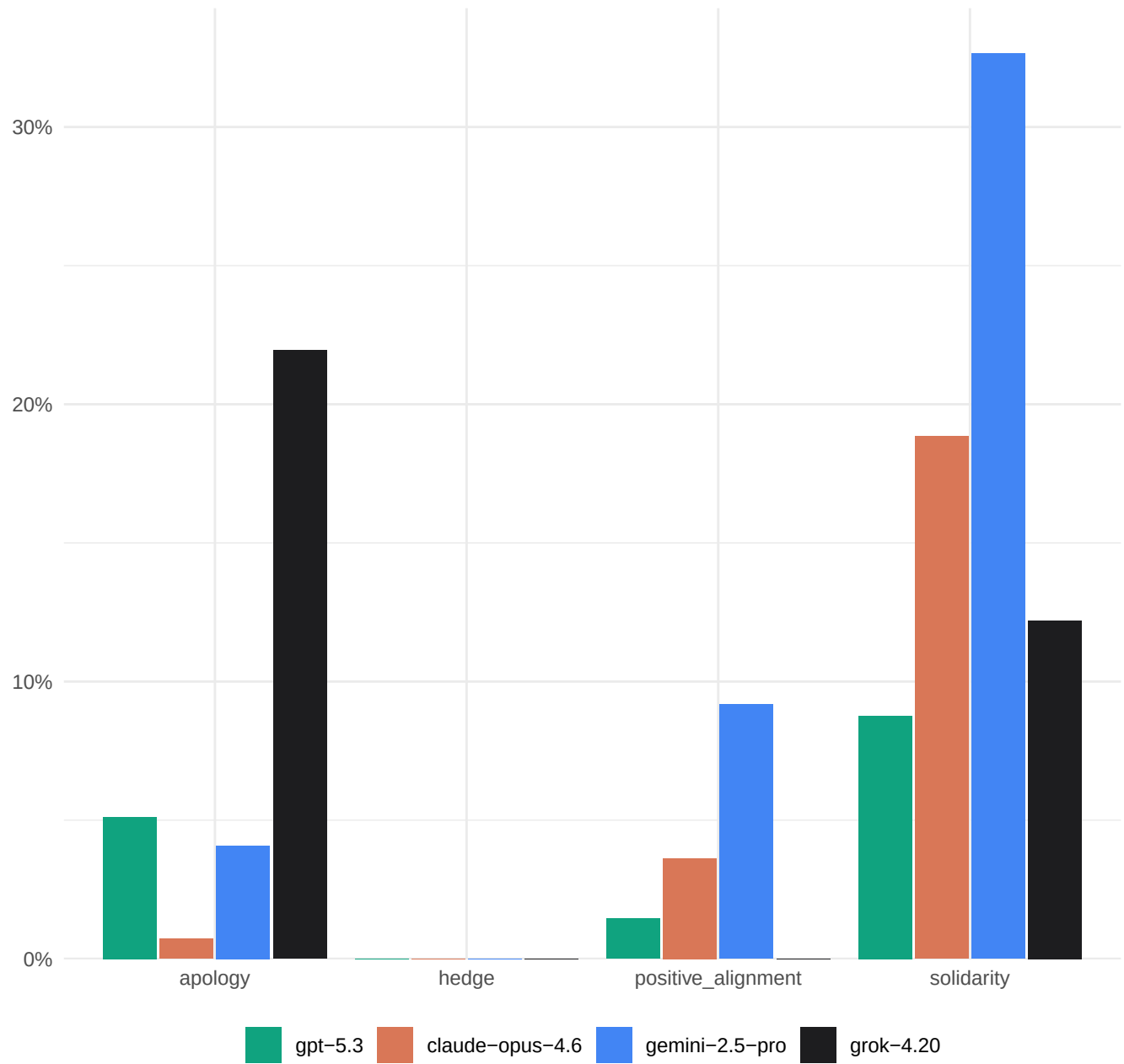


Flagship × L2 feature × harm category

Rate within each (model, category) NC set. Cells with n < 3 masked.

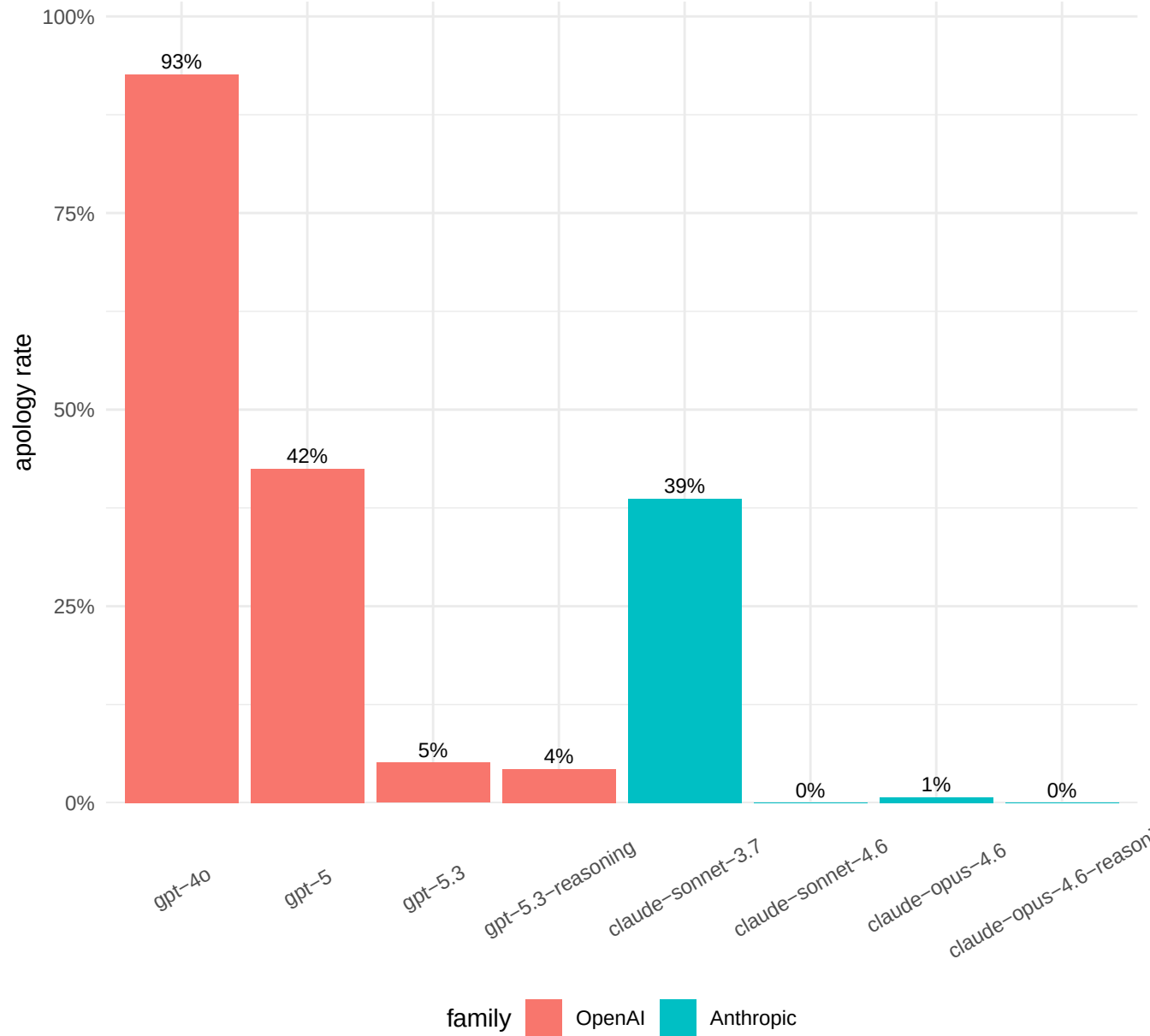


Flagship affiliative / mitigating features



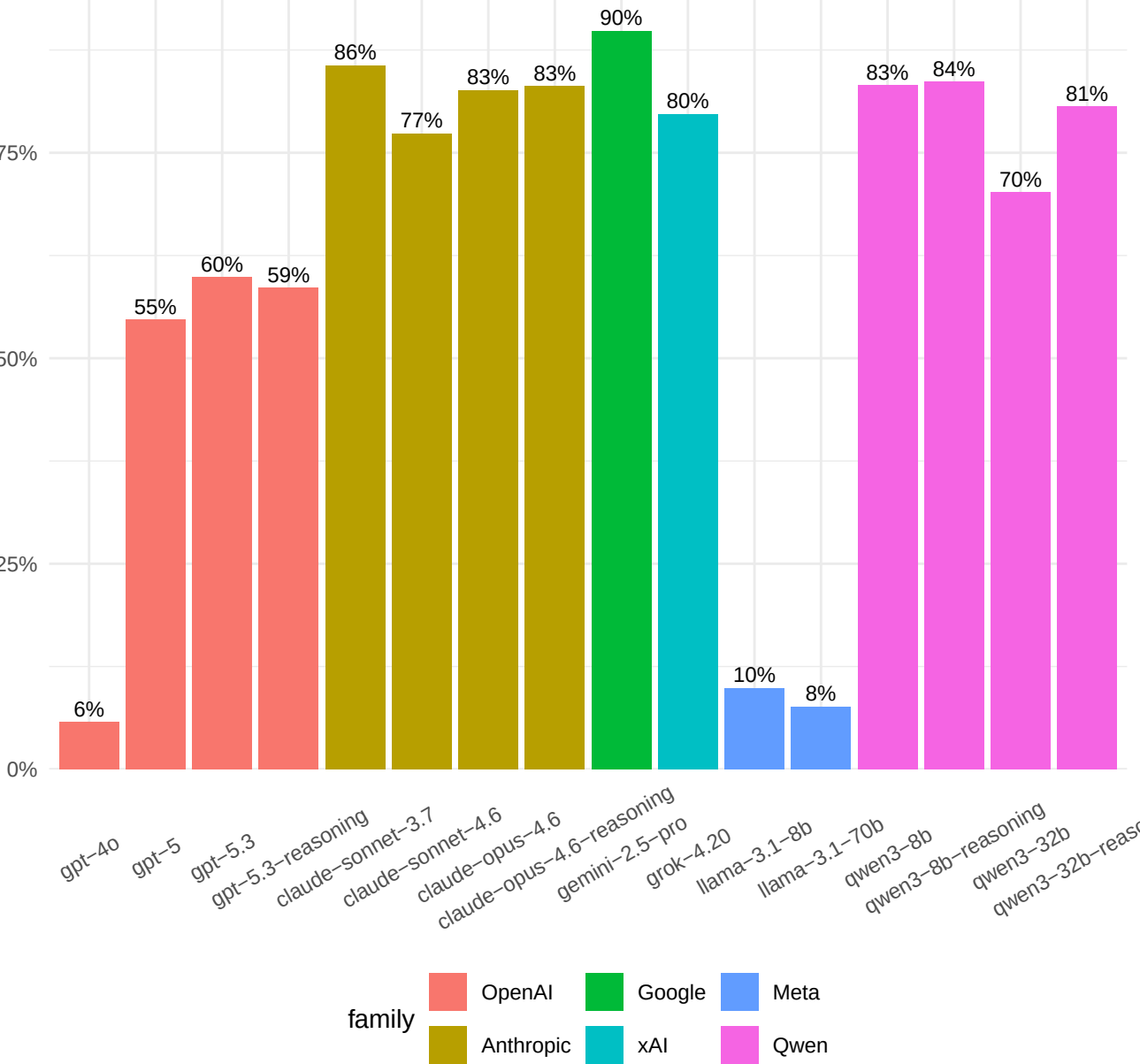
Apology in non-compliance: OpenAI vs Anthropic

Earlier models lean on apology; newer ones do not



Negative stance prevalence by model

negative_stance_rate



Affiliative refusal features by model

